

HASSAN GHANAM

Software Engineer | Backend Developer

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PROFESSIONAL SUMMARY

Fresh Graduate Software & Information Systems Engineer from Damascus University with expertise in backend engineering, clean architecture, and distributed systems. Proficient in building scalable RESTful APIs using PHP/Laravel with SOLID design principles, payment gateway integration (Stripe), and Firebase. Experienced in Search & Information Retrieval Systems, Vector Databases (FAISS), Advanced Algorithms (Minimax, BFS, DFS, A*.....), Compiler Design, Asynchronous Queues, and Redis Caching. Proven track record in competitive programming (Codeforces Pupil, DCBC participant) with solid algorithmic problem-solving skills.

TECHNICAL SKILLS

- Backend & API Development: PHP, Laravel (RESTful APIs, MVC, DTOs, Repositories, Events & Listeners, Custom Middleware), Python (Flask, Blueprints), Payment Integration (Stripe), Firebase, Postman API, JSON, C++, Java, JavaScript, HTML/CSS.
- Software Architecture & Modeling: Design Patterns & Anti-Patterns, Component-Based Software Engineering (CBSE), Service-Oriented Architecture (SOA), Clean Architecture, SOLID Principles, Real-time Systems & Petri Nets, Meta-Modeling, SDLC, UML.
- Algorithms & Game : Minimax Algorithm (Alpha-Beta Pruning), Graph Traversal (BFS, DFS), Heuristic Search, State-Space Exploration, Game Logic Optimization, Compiler Design (Lexer, Parser, AST, Code Gen).
- Database & Data Engineering: MySQL, SQLite, SQL Server, PL/SQL, Relational Databases, ERD Design, Data Warehousing, Big Data Lifecycle, ETL Processes, Star Schema & Dimensional Modeling (SCD), MapReduce & Hadoop Concepts, OLAP, ACID Compliance.
- System Infrastructure & Performance: Asynchronous Queues & Jobs, Caching & Session Storage (Redis), Connection Pooling, Rate Limiting, Apache, VMware.
- Distributed Systems & Cloud: Microservices vs Modular Monolith, Event-Driven Architecture (Apache Kafka, RabbitMQ), Saga & Outbox Patterns, Load Balancing, Fault Tolerance, Cloud Computing Models (IaaS, PaaS, SaaS).
- Information Retrieval & Search Mechanics: BM25 Fine-Tuning, Vector Space Model (VSM / TF-IDF), Semantic Search & Embeddings (Sentence-Transformers / BERT), Vector Database Integration (FAISS / CPU Indexing), Text Preprocessing (NLTK, Tokenization, Stemming vs Lemmatization), Hybrid Search (Parallel & Serial Fusion), IR Metrics (MAP, NDCG, Precision@K).
- Tools & Methodologies: Git, GitHub, Agile / Scrum , JIRA, Requirements Gathering & Analysis, Automated CI/CD Pipelines, Data Structures & Algorithms.

KEY PROJECTS & EXPERIENCE

Unified Digital Portal for Syrian Universities — Graduation Project

Dec 2025 – Aug 2026

Laravel, PHP, Redis, MySQL, RESTful APIs, CI/CD

- Architected and developed the backend for a unified digital portal automating services across Syrian universities.
- Implemented Clean Architecture using DTOs, Repository Contracts, Service Layers, Enums, Policies, and Custom Responses.
- Integrated Laravel Jobs & Listeners with Redis queues for asynchronous event processing, alongside Database Connection Pooling, Rate Limiting middleware, and Automated CI/CD pipelines.

Information Retrieval Engine (IR-System)

Jun 2026 – Jun 2026

Python, Flask, FAISS, NLTK, Sentence-Transformers, SQLite

- Designed and engineered a Service-Oriented Architecture (SOA) modular search engine processing 382k+ documents (BEIR Touche-2020 dataset).
- Implemented and benchmarked 4 retrieval models: Sparse Vector Space Model (TF-IDF), Probabilistic BM25, Deep Semantic Embeddings (BERT/all-MiniLM-L6-v2), and Hybrid CombSUM/Serial RRF Search models.
- Fine-tuned BM25 hyperparameters ($k_1=2.0$, $b=0.60$) achieving peak performance at $\text{MAP}@10 = 0.1470$ and $\text{Precision}@10 = 76.12\%$.
- Integrated FAISS vector store with L2 normalization and memory-mapped files (mmap, float32), reducing BERT query latency from ~1.5s to ~0.1s.
- Built full preprocessing pipelines (NLTK stemming, Unicode normalization, field-weighted titling) alongside a dynamic web analytics interface (Chart.js, Flask Blueprints).

Complaints Management System

Nov 2025 – Dec 2025

Laravel, PHP, Redis, MySQL, RESTful APIs

- Developed a scalable backend system for handling citizen complaints and municipal requests, applying high-performance web architecture concepts.
- Leveraged Redis caching, session management, message queues, rate-limiting protocols, and audit tracing for high reliability and availability.

Compiler Design Project

May 2025 – Sep 2025

Custom ANTLR-to-Web Transpiler, Java, Parsing & Compilation Tools

- Built a full multi-stage compiler that converts custom ANTLR-like grammar specifications directly into executable HTML, CSS, and JavaScript for browser environments.
- Designed and implemented all compilation pipeline stages: Lexical Analysis (Lexer), Parsing (AST construction), Semantic Analysis, and Target Code Generation.

MatchLab – Medical Laboratories Management System

Jun 2025 – Oct 2025

Laravel, PHP, MySQL, RESTful APIs

- Engineered a multi-actor backend platform for medical analysis laboratories to manage patient records, test results, and lab workflow statuses.
- Implemented role-based access control (RBAC) to manage multi-tier user roles (Patients, Technicians, Admins) and secure sensitive medical data.

Ludo Star Console Game – AI & Graphic Rendering

Jun 2025 – Jul 2025

Java, OOP, Game AI, Decision Trees

- Engineered a multi-mode CLI implementation of Ludo Star (Player vs Player & Player vs AI) featuring dynamic ANSI color-rendered board state updates.
- Implemented Expectimax (ExpVal) decision-making algorithm integrated with Alpha-Beta pruning to model probabilistic dice outcomes and optimal strategy.
- Applied Object-Oriented Programming (OOP) principles to structure board mechanics, player states, turn validation, and state-space tree traversals.

Light Link Mirror – Algorithmic Puzzle & Solver Game

May 2025 – Jun 2025

Java, OOP, State-Space Search, Algorithms

- Built a single-player CLI puzzle game featuring automated optical ray-casting and board state-space rendering.
- Engineered an AI pathfinder/solver utilizing state evaluation logic and heuristic graph search to calculate optimal reflection trajectories and solve levels automatically.
- Applied Object-Oriented Design to dynamic grid management, mirror rotation mechanics, and light-beam path verification.

TRAVANIX — Tourism Management System

May 2024 – Aug 2024

Laravel, PHP, MySQL, RESTful APIs, Map Integration

- Built a multi-featured backend tourism platform enabling users to browse curated trips, manage composite bookings (entire trips, individual hotel rooms, or activities), and track real-time booking status workflows with admin orchestration.
- Implemented comprehensive account management modules featuring dynamic trip/venue filtering, favorites tracking, user ratings, and nested comment systems.
- Integrated location-based mapping features to deliver enhanced navigation for hotels, restaurants, and activity destinations.

Medixify — Pharmacy & Warehouse Management System

Dec 2023 – Jan 2024

Laravel, PHP, MySQL, RESTful APIs

- Designed and developed a comprehensive pharmaceutical ecosystem connecting medicine suppliers/warehouses with retail pharmacies to streamline drug distribution and order management.
- Implemented core inventory operations including dynamic batch tracking, medicine search/filtering, automated stock calculations, and real-time order processing workflows.
- Built role-specific authorization layers for warehouse managers and pharmacists to ensure secure transaction management and stock accuracy.

EDUCATION

Bachelor's Degree in Information Technology Engineering

Sep 2021 – Sep 2026

Specialization in Software Engineering & Information Systems

Damascus University, Syria

ACHIEVEMENTS & COMPETITIONS

- Codeforces: Achieved Pupil rank through continuous competitive programming and algorithmic problem solving.  **Codeforces: HASSAN_GH**
- Damascus Collegiate Programming Contest (DCPC): Active participant in the official university competitive programming contest.

LANGUAGES

- Arabic: Native
- English: Professional Working Proficiency